SOLUTIONS: COMPLEX CALCULUS
EE2L

1. Exercise

a) (similar to examples done in class) The curve $\partial^+ B$ can be parameterized as γ given below:

$$\gamma(t) = e^{it}, \qquad t \in [0, 2\pi].$$

By definition of integral of complex variable we have:

$$\int_{\partial^+ B} \frac{e^{iz}}{z}\, dz = \int_0^{2\pi} \frac{e^{i\gamma(t)}}{\gamma(t)} \dot{\gamma}(t)\, dt.$$

[2]

Since $\dot{\gamma}(t) = ie^{it}$ we see that:

$$\int_{\partial^+ B} \frac{e^{iz}}{z}\, dz = \int_0^{2\pi} ie^{i(\cos(t)+i\sin(t))}\, dt =$$

$$= \int_0^{2\pi} ie^{-\sin(t)} e^{i\cos(t)}\, dt =$$

$$= \int_0^{2\pi} ie^{-\sin(t)} [\cos(\cos(t)) + i\sin(\cos(t))]\, dt.$$

This proves that:

$$I = \mathrm{Im}\left[\int_{\partial^+ B} \frac{e^{iz}}{z}\, dz \right].$$

[2]

Common mistake: $\mathrm{Im}[e^{iz}] = \sin(z)$. This formula only holds for $z \in \mathbb{R}$. In fact, while: $e^{iz} = \cos(z) + i\sin(z)$ is always true, one should not forget that for $z \in \mathbb{C}$ both $\cos(z)$ and $\sin(z)$ are complex numbers, and therefore the imaginary part should be the sum of the real part of $\sin(z)$ and the imaginary part of $\cos(z)$.

b) (similar to results derived in class) The function e^{iz} is a composition of holomorphic functions (multiplication times complex scalar and complex exponential). It is therefore holomorphic in $\mathbb{C}$. The function z is also holomorphic in $\mathbb{C}$, therefore their ratio is holomorphic whenever the denominator is different from 0. This is also the domain of definition of the function. The point $z = 0$ is the only singularity, and it is a pole of multiplicity 1, as it is easily verified by noticing that $e^{i0} = 1 \neq 0$ and realizing that $z \cdot \frac{e^{iz}}{z} = e^{iz}$ for all $z \neq 0$, which, as previously claimed is holomorphic. [3]

c) (similar to examples done in class) The residue of the function at 0 is given by:

$$\lim_{z \to 0} z \frac{e^{iz}}{z} = e^{i0} = 1.$$

Hence, by the Residue's theorem:

$$\int_{\partial^+ B} \frac{e^{iz}}{z}\, dz = 2\pi i \cdot 1 = 2\pi i.$$

It follows that $I = 2\pi$. [3]

Common mistake: trying to use a semicircle to find the integral. The semicircles are used to find improper integrals on the real line by making sure that the semicircle at infinity yields a 0 contribution (using Jordan's lemma or similar tools). This is not needed here as the integral is taken over the unit circle in complex plane.

d) (similar to problems solved in class) Differentiating u with respect to x and y yields:

$$\frac{\partial u}{\partial x} = \mathrm{e}^{x}\cos(y) + \mathrm{e}^{-y}\cos(x),$$
$$\frac{\partial u}{\partial y} = -\mathrm{e}^{x}\sin(y) - \mathrm{e}^{-y}\sin(x).$$

[1]

Differentiating again, yields:

$$\frac{\partial^2 u}{\partial x^2} = \mathrm{e}^{x}\cos(y) - \mathrm{e}^{-y}\sin(x),$$
$$\frac{\partial^2 u}{\partial y^2} = \mathrm{e}^{-y}\sin(x) - \mathrm{e}^{x}\cos(y).$$

[2]

It is easy to see that:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,$$

which proves that u is a solution of the Laplace's equation in $\mathbb{R}^2$ and hence harmonic. [1]

e) (similar to problems solved in class) To find the conjugate v we exploit the Cauchy-Riemann's equations. In particular:

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = \mathrm{e}^{x}\cos(y) + \mathrm{e}^{-y}\cos(x).$$

This implies:

$$v(x,y) = \int \mathrm{e}^{x}\cos(y) + \mathrm{e}^{-y}\cos(x)\, dy =$$
$$= \mathrm{e}^{x}\sin(y) - \mathrm{e}^{-y}\cos(x) + c_1(x),$$

where $c_1(x)$ is an arbitrary function of x. [1]

Similarly:

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = \mathrm{e}^{x}\sin(y) + \mathrm{e}^{-y}\sin(x).$$

This implies:

$$v(x,y) = \int \mathrm{e}^{x}\sin(y) + \mathrm{e}^{-y}\sin(x)\, dx =$$
$$= \mathrm{e}^{x}\sin(y) - \mathrm{e}^{-y}\cos(x) + c_2(y),$$

where c_2 is an arbitrary function of y. [1]

Equating the previous expressions for v yields:

$$c_1(x) = c_2(y) = c$$

for some scalar $c \in \mathbb{R}$. Hence:

$$v(x,y) = \mathrm{e}^{x}\sin(y) - \mathrm{e}^{-y}\cos(x) + c.$$

[2]

f) (similar to examples done in class) To find the function g as requested we evaluate $u+iv$ on the real axis ($y=0$). This gives:

$$u(x,0)+iv(x,0)=e^x+\sin(x)+i[-\cos(x)+c].$$

Hence we may let:

$$g(z)=e^z+\sin(z)-i\cos(z)+ic=e^z-ie^{iz}+ic.$$

The function $g(z)$ equals $u(x,y)+iv(x,y)$ on the real axis, and is clearly holomorphic. Hence it coincides with $u(x,y)+iv(x,y)$ for all $(x,y)\in\mathbb{R}^2$. [3]

g) (unseen) A holomorphic function g is conformal at every point where $g'(z)\neq 0$. Hence, provided z is such that:

$$g'(z)=e^z-i^2e^{iz}=e^z+e^{iz}\neq 0$$

[1]

Notice that, to have

$$e^z=-e^{iz}=e^{i(z+\pi)}$$

we need:

$$\mathrm{Re}(z)=\mathrm{Re}(i(z+\pi))=-\mathrm{Im}(z)$$

(which implies the two sides of the previous equation have the same modulus), and

$$\mathrm{Im}(z)+2k\pi=\mathrm{Im}(i(z+\pi))$$

(which implies the two sides of the previous equation have the same argument). [2]

We have a (linear) system of two equations in two real unknowns, $\mathrm{Re}(z)$, $\mathrm{Im}(z)$, whose solution is:

$$-\mathrm{Re}(z)=\mathrm{Im}(z)=\frac{2k-1}{2}\pi,$$

with k an arbitrary integer. Hence there are infinitely many point in $\mathbb{C}$ where g is not conformal, these are:

$$z=-\frac{2k-1}{2}\pi+i\frac{2k-1}{2}\pi,\qquad k\in\mathbb{Z}.$$

[1]

Common mistake: e^z+e^{iz} is always different from zero. Or, e^z+e^{iz} is always positive. Notice that complex exponentials are complex numbers, and therefore they do not have a sign. Their sum can be zero even if both of them have modulus different from zero.

2. Exercise (Similar to equations solved in class)

a) Let $X(s)$ denote $\mathscr{L}[x]$. Then $\mathscr{L}[dx/dt]=sX(s)$ and $\mathscr{L}[d^3x/dt^3]=s^3X(s)$. Applying Laplace's transforms to both sides of the differential equation yields:

$$(s^3+3s-4)X(s)=\frac{1}{s}.$$

[6]

Common mistake: $\mathscr{L}[u(t)]=1$; the latter is the Laplace transform of the Dirac's delta function, not of the unit step function

Solving with respect to $X(s)$ yields:

$$X(s) = \frac{1}{s(s^3+3s-4)}.$$

[4]

Notice that: $(s^3+3s-4) = (s-1)(s^2+s+4)$, therefore:

$$X(s) = \frac{A}{s-1} + \frac{Bs+C}{s^2+s+4} + \frac{D}{s}$$

with:

$$A = \lim_{s\to 1}(s-1)X(s) = \frac{1}{6}$$

$$D = \lim_{s\to 0} sX(s) = -\frac{1}{4}$$

$$C = -\frac{1}{12}, \qquad B = \frac{1}{12}.$$

[5]

Common mistake: use a decomposition where only the term $\frac{C}{s^2+s+4}$ appears, and therefore B is automatically set to 0. In general, both terms will be needed.

Taking inverse Laplace's transforms yields:

$$x(t) = \frac{e^t}{6} + \frac{e^{-\frac{t}{2}}\left(\cos\left(\frac{\sqrt{15}t}{2}\right) - \frac{\sqrt{15}\sin\left(\frac{\sqrt{15}t}{2}\right)}{5}\right)}{12} - \frac{1}{4}.$$

[5]

b) Since $x(t)$ needs to be constant we do have $\dot{x}(t) \equiv 0$, $\ddot{x}(t)$ and $x^{(3)}(t) \equiv 0$ for all $t > 0$. In particular, then, $-4x(t) = 1$ for all $t > 0$. Since $\dot{x}$, $\ddot{x}$ and x are continuous in t we may pick:

$$x(0) = -\frac{1}{4},\ \dot{x}(0) = 0,\ \ddot{x}(0) = 0.$$

[5]

Common mistake: not realizing that $x(t)$ constant means $X(s) = x(0)/s$ and not $1/s$.